

Event Organization Smart Contract Project (Blockchain and Cryptocurrency)

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Introduction

Overview of the Project

01

Purpose

The Event Org. Smart Contract Project aims to revolutionize event management by leveraging blockchain technology and smart contracts.

02

Benefits

By automating event processes and transactions, the project enhances efficiency, transparency, and security for event organizers and attendees.

SMART CONTRACTS

Technical article and infographic about theory and development of smart contracts

WHAT ARE SMART CONTRACTS?

A smart contract is a computer protocol designed to digitally facilitate, validate or negotiate or execute a contract.



MAIN FEATURES

Automatically executable

No person is required to oversee the execution of the smart contract. If the conditions are met, the contract will be executed automatically.

Enforcable

The contract must be described in such a way that it can be implemented from a legal and infrastructural point of view.

Semantically sound

The language in which the smart contract is written must be clear to the machine and to the people!

HISTORY OF SMART CONTRACTS

1997

Nick Szabo - the first scientific paper to mention smart contracts

2008

The first smart contracts were implemented in the Bitcoin blockchain

2014

Invention of Ethereum, a platform specialising in smart contracts

SOME USE CASES OF SMART CONTRACTS

Healthcare

Realestate

Elections

Made with PopAi

Contract Features.

Creation of Events

The smart contract enables the seamless creation of events, including defining event details, ticket availability, and pricing.

Purchase of Tickets

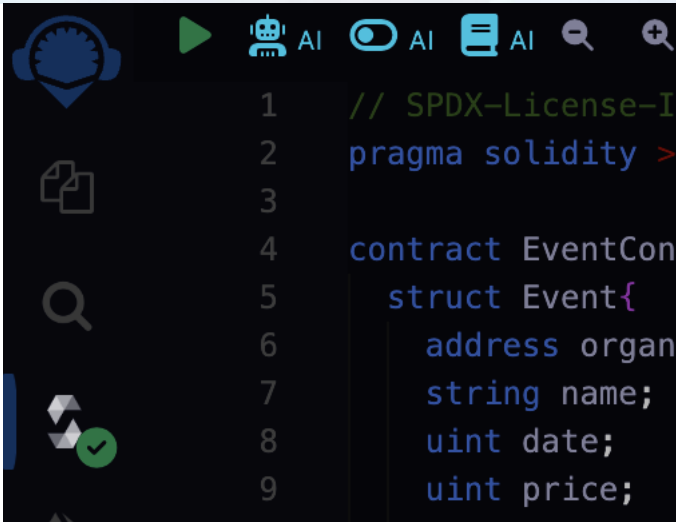
Attendees can securely purchase event tickets through the smart contract, ensuring transparency and authenticity.

Transfer of Tickets

The smart contract facilitates the transfer of tickets between users, providing a decentralized and tamper-proof ticketing system.

Smart Contract Functionality

Using Remix for Development



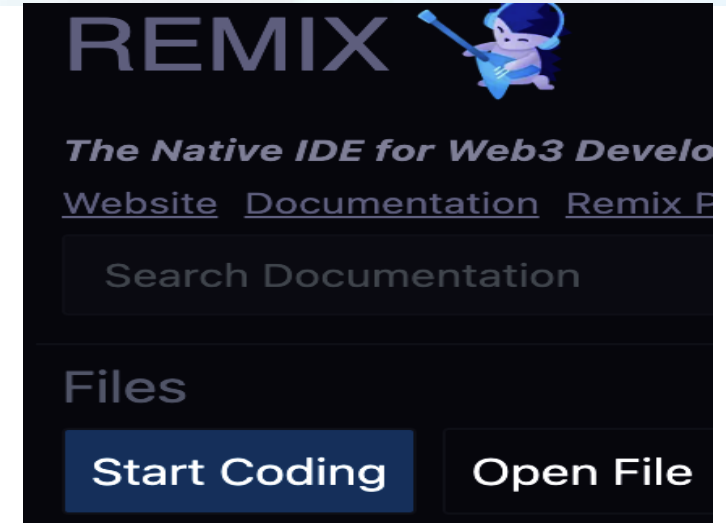
Introduction to Remix IDE

Remix is a web-based integrated development environment that simplifies the writing, testing, and deployment of smart contracts.



Features and Benefits

Offers a user-friendly interface, syntax highlighting, and a built-in Solidity compiler.



Setting up the Env.

Developers can easily set up the Remix environment for smart contract development, making it accessible and convenient.

Overview Sample Code Snippets

Examples of Solidity code snippets for creating events, buying tickets, and transferring tickets, showcasing the simplicity and effectiveness of smart contract development.

Overview Sample Code Snippets (Cont')

```
function sellTicket(uint id, uint quantity) public {    ⛽ infinite gas
    require(events[id].date != 0, "Event Does Not Exist");
    require(events[id].date > block.timestamp, "Event has ended");
    require(tickets[msg.sender][id] >= quantity, "You do not have enough tickets to sell");
    Event storage _event = events[id];
    _event.ticketRemaining += quantity;
    tickets[msg.sender][id] -= quantity;
    // Transfer ether to the seller
    uint refundAmount = _event.price * quantity;
    payable(msg.sender).transfer(refundAmount);
}
```

```
function transferTicket(uint id, uint quantity, address to) public {    ⛽ infinite gas
    require(events[id].date != 0, "Events Does Not Exist");
    require(events[id].date > block.timestamp, "Event has ended");
    require(tickets[msg.sender][id] >= quantity, "You do not have tickets to transfer");
    tickets[msg.sender][id] -= quantity;
    tickets[to][id] += quantity;
}
```

Overview Sample Code Snippets (Cont')

```
function createEvent(string calldata name,uint date,uint price,uint ticketCount) public{
    require(block.timestamp<date,"You cannot create an event for past date");
    require(ticketCount>0,"Ticket count must be greater than 0");
    events[nextId]=Event(msg.sender, name,date,price,ticketCount,ticketCount);
    nextId++;
}
```

```
✓ function buyTicket(uint id, uint quantity) public payable{    infinite
    require(events[id].date!=0,"Events Does Not Exist");
    require(events[id].date>block.timestamp,"Event has ended");
    Event storage _event=events[id];
    require(msg.value==(_event.price*quantity),"Ether is not enough");
    require(_event.ticketRemaining>=quantity,"Not enough Tickets left");
    _event.ticketRemaining-=quantity;
    tickets[msg.sender][id]+=quantity;
```

Remix Development

Terms:

Gas limit:

This sets the maximum amount of gas that will be allowed for all the transactions created in Remix.

Value field:

This sets the amount of ETH, WEI, that is sent to a contract or a payable function.

At Address:

At Address is used to access a contract that has already been deployed. Because the contract is already deployed, accessing a contract with At Address does not cost gas.

Deployment

createEvent

name:	AB
date:	1715249443
price:	10
ticketCount:	100

 Calldata  Parameters

Create Event

A step-by-step guide on deploying the smart contract on the blockchain, highlighting the simplicity and accessibility of the deployment process.

buyTicket

id:	1
quantity:	50

 Calldata  Parameters

Buy Ticket

Instructions for users on how to create events, buy tickets, and transfer tickets post-deployment, ensuring a seamless user experience.

transferTicket

id:	1
quantity:	25
to:	0x4B20993Bc481177

 Calldata  Parameters

Transfer Ticket

A step-by-step guide on deploying the smart contract on the blockchain, highlighting the simplicity and accessibility of the deployment process.

Conclusion

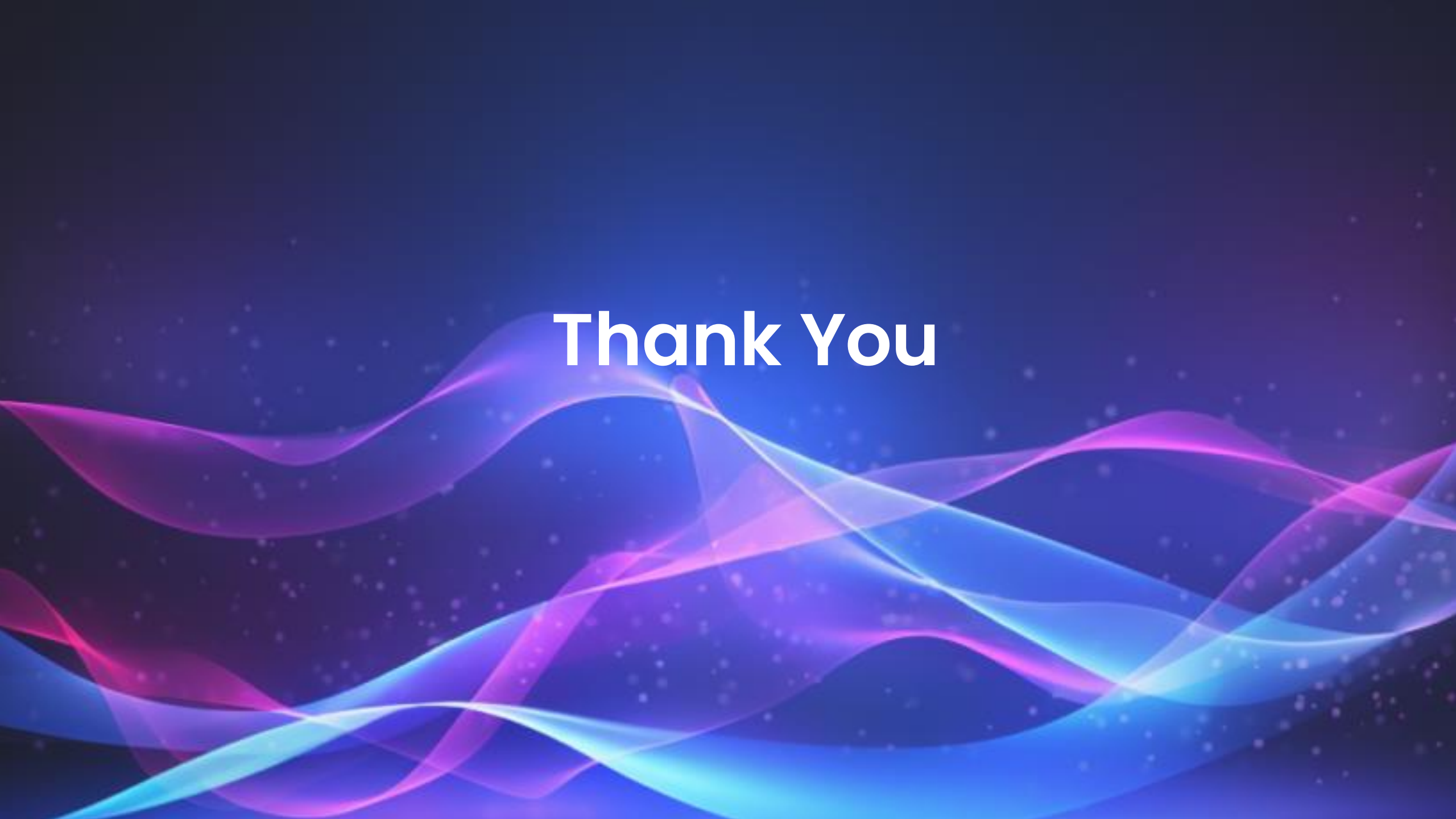
Conclusion:

The Event Organization Smart Contract Project revolutionizes event planning by automating tasks through blockchain technology.

Smart contracts ensure transparency, security, and efficiency in ticket sales, venue booking, and vendor payments.

By eliminating intermediaries, the project promotes trust and accountability, enhancing the event experience for all stakeholders.

Thank You

The background features a dark blue gradient with several translucent, flowing ribbons in shades of blue and purple. These ribbons create a sense of movement and depth. Scattered throughout the background are numerous small, white, star-like specks, giving the overall image a cosmic or ethereal feel.